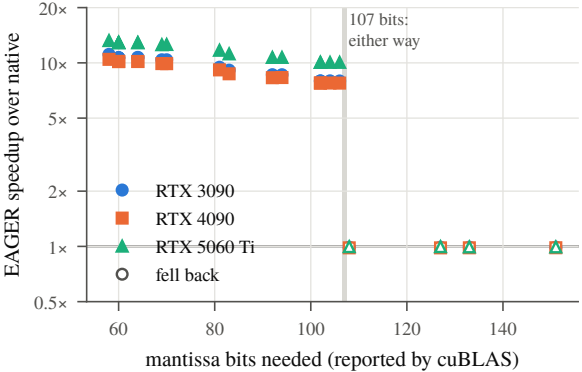
(a) bits needed; shaded = runtime fallback to native

k (m = n = 4096)	7.25	7.5	7.6	7.7	7.75	7.8	7.9	8
16k	103	103	105	105	105	105	**107**	**107**
8k	103	103	105	105	105	105	**107**	**107**
4k	104	104	106	106	106	106	108	108
2k	104	105	106	**107**	**107**	**107**	108	108
1k	104	105	**107**	**107**	**107**	**107**	109	109
512	104	106	**107**	**107**	108	108	109	109
256	106	**107**	109	109	109	109	110	111

emulated
fell back

φ (exponent range inside the dot products)